

QIFAN ZHANG

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WORK EXPERIENCE

Palo Alto Networks
Senior Staff Researcher

Jan 6, 2025 - current

EDUCATION

University of California, Irvine
Ph.D. in Computer Engineering

Sept 2020 - Mar 2025

Advisor: Prof. Zhou Li

Department of EECS, the Henry Samueli School of Engineering

University of California, Irvine

Sept 2020 - Jun 2022

Master of Science in Computer Engineering

Advisor: Prof. Zhou Li

Department of EECS, the Henry Samueli School of Engineering

ShanghaiTech University

Aug 2016 - Jul 2020

B.Eng. in Computer Science and Technology

Minor in Innovation and Entrepreneurship

University of California, Berkeley

2017

Summer Session

RESEARCH INTERESTS

Domain Name System (DNS). I study DNS security, privacy, and reliability, with a focus on automated vulnerability discovery through fuzzing and formal analysis. My work includes ResolverFuzz [Security'24], DNSSECVerif [CCS'26], Phoenix Domain [NDSS'23], and MaginotDNS [Security'23].

Machine Learning Security and Privacy. I'm also interested in security and privacy topics related to machine learning. My past research demonstrated model extraction on Autonomous Vehicle using Gradient-Descent based methods [ACSAC'22]. Recently, I participated in FedMLSecurity [KDD'24], a benchmark to simulate attacks and defenses on Federated Learning and Large Language Models (LLMs), and CustodianFL [PRISM'26] for verifiable anomaly detection in federated learning.

SELECTED PUBLICATIONS

Conference Papers

- **Zhang, Q.**, Shen, Z., Karim, I., Bertino, E., Liu, D. and Li, Z. *Mind the Gap: Cross-RFC Formal Analysis Exposes DNSSEC's Hidden Vulnerabilities*. Accepted by the ACM SIGSAC Conference on Computer and Communications Security (**CCS**), 2026. DNSSECVerif: GitHub Artifacts. Paper and code under embargo.
- **Zhang, Q.**, Bai, X., Li, X., Duan, H., Li, Q. and Li, Z. *ResolverFuzz: Automated Discovery of DNS Resolver Vulnerabilities with Query-Response Fuzzing*. Accepted by the 33rd USENIX Security (**Security**), 2024. Extended version available on ArXiv. Artifacts available on GitHub.
 - **12** types of vulnerabilities, **23** bugs detected and **15** CVEs assigned among 6 popular DNS software.
 - Presented in DNS-OARC'42 and NDSS 2024 poster session.
 - **Skills involved:** CVE reading and summary, Grammar-based fuzzing, Network environment settings on Docker, Code analysis on DNS software, Cloudflare API, concurrent programming.

- **Zhang, Q.**, Shen, J., Tan, M., Zhou, Z., Li, Z., Chen, Q.A. and Zhang, H. *Play the Imitation Game: Model Extraction Attack against Autonomous Driving Localization*. Accepted by the 38th Annual Computer Security Applications Conference (**ACSAC**), 2022.
 - Achieve cm-level precision with 40-second driving data.
 - **Skills involved:** model establishment and training on PyTorch, Optimization, Baidu Apollo, Autonomous Driving controller algorithms.
- Han, S., Buyukates, B., Hu, Z., Jin, H., Jin, W., Sun, L., Wang, X., Xie, C., Zhang, K., **Zhang, Q.**, Zhang, Y., Avestimehr, S. and He, C. *FedMLSecurity: A Benchmark for Attacks and Defenses in Federated Learning and LLMs*. Accepted by ACM Knowledge Discovery and Data Mining Conference (**KDD**), 2024. Preprint available on arXiv. Artifacts available on GitHub.
- Li, X., Lu, C., Liu, B., **Zhang, Q.**, Li, Z., Duan, H. and Li, Q. *The Maginot Line: Attacking the Boundary of DNS Caching Protection*. Accepted by the 32nd USENIX Security (**Security**), 2023.
 - **Vulnerability acknowledged** by CVE-2021-25220 (BIND 9), CVE-2021-43105 (Technitium), CVE-2022-32983 (Knot Resolver).
 - Awarded \$1,000 by Microsoft Security Response Center.
 - **Skills involved:** Network environment settings on Virtual Machine, debugging via GDB and CLion, Python Scapy, Code analysis on DNS software.
- Li, X., Liu, B., Bai, X., Zhang, M., **Zhang, Q.**, Li, Z., Duan, H. and Li, Q. *Ghost Domain Reloaded: Vulnerable Links in Domain Name Delegation and Revocation*. Accepted by the 30th Annual Network and Distributed System Security Symposium (**NDSS**), 2023.
 - **Vulnerability acknowledged** by CVE-2022-30250, CVE-2022-30251 (Knot Resolver), CVE-2022-30252 (PowerDNS Recursor), CVE-2022-30254 (Simple DNS Plus), CVE-2022-30256 (MaraDNS), CVE-2022-30257, CVE-2022-30258 (Technitium), CVE-2022-30698, CVE-2022-30699 (Unbound)
 - **Skills involved:** Network scanning and measurement, Network environment settings on Docker, Python Scapy, Code analysis on DNS software.

Workshop Papers

- Han, S., Wu, W., Buyukates, B., Jin, W., **Zhang, Q.**, Yao, Y. and Avestimehr, S. *Kick Bad Guys Out! Conditionally Activated Anomaly Detection in Federated Learning with Zero-Knowledge Proof Verification*. Published in the Workshop on Attack Provenance, Reasoning, and Investigation for Security in the Monitored Environment (**PRISM**), co-located with NDSS 2026. DOI.

SERVICES

Program Committee

- USENIX Security: 2027
- ACSAC: 2026, 2025
- SiMLA: 2025

Artifact Evaluation Committee

- CCS: 2024, 2023
- USENIX Security: 2024
- NDSS: 2025, 2024

External Reviewers

- NDSS: 2025, 2023, 2022, 2021
- AsiaCCS: 2022, 2021
- SecureComm: 2023
- IEEE Transactions on Dependable and Secure Computing (TDSC)

- IEEE Transactions on Information Forensics & Security (T-IFS)
- IEEE Transactions on Wireless Communications (TWC)
- IEEE Internet of Things Journal (IoTJ)
- Elsevier Computer Networks
- Elsevier High-Confidence Computing
- Springer Peer-to-Peer Networking and Applications (PPNA)
- PeerJ Computer Science

TECHNICAL SKILLS

Programming Language	Python, Java, C/C++, Rust
Software & Tools	Matlab/Simulink, VMware Workstation Player, Docker, Cloudflare API, OpenCV, CLion, GDB

TEACHING

Teaching Assistant University of California, Irvine

- (Head TA) EECS 148 (S24): Computer Networks (#students: 217)
- (Head TA in F23) EECS 40 (F23, F22): Object-Oriented System & Programming (#students: 90/95)

Teaching Assistant ShanghaiTech University

- (core TA) SI 100B (S18, S19, S20): Intro. to Info. Science and Technology (#students: 203/174/410)
- CS 100 (F18): Programming (#students: 243)
- CS 277 (F19): Intro. to Data Science and FinTech (#students: 23)
- SI 100C (F17): Intro. to Computer Science and Technology (#students: 127)

HONORS

University of California, Irvine

- Student travel grant for USENIX Security (2024, 2021)
- Associated Graduate Students Conference Stipend (Winter 2024, Winter 2022)
- 2024 UCI Concerto Competition Winner
- 2023 ANRW Travel Grant
- ACSAC Student Conferenceship (Fall 2022)
- Student travel grant for NDSS (2021)
- Student travel grant for IEEE Symposium on Security and Privacy (2021)

ShanghaiTech University

- 2020 ShanghaiTech Outstanding Graduate
- SIST Outstanding Teaching Assistant (2020, 2019, 2018)
- Merit Student (2018-2019, 2017-2018, 2016-2017)
- Outstanding Personnel in 2017 Summer Camp